

DATA/CS 172: Python for Data Analysis

Intro to Lists

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9/16/2024



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Reminder

- Friday's lecture (recording, slides, demo) is up on Laulima
- Lab 5 due today

Agenda

- List Basics
- Lists and Strings

Storing data

The diagram shows the Drake Equation: $N = R^* \times f_p \times \eta_e \times f_i \times f_c \times L$. Each variable is labeled with a handwritten note and an arrow pointing to it:

- R^* : The average rate of star formation per year in our galaxy (indicated by two stars).
- f_p : The fraction of stars with planets (indicated by a planet icon).
- η_e : The average number of planets that can potentially support life (per star with planets).
- f_i : The fraction that can go on to support intelligent life (indicated by a brain icon).
- f_c : The fraction of civilizations that develop a technology detectable from space (indicated by a radio dish icon).
- L : Length of time such civilizations release detectable signs into space (indicated by a signal wave icon).

The entire diagram is captioned "The Drake Equation."

- We know how we can read a file containing multiple lines of data...
- Great! But so far, no where to store that data!
- What if I need to store 100 pieces of data in Python?
 - Could use 100 variables (Ugh!)
 - Could store them in a giant string (slow & painful)
 - Could use a file for storage (slow & painful)

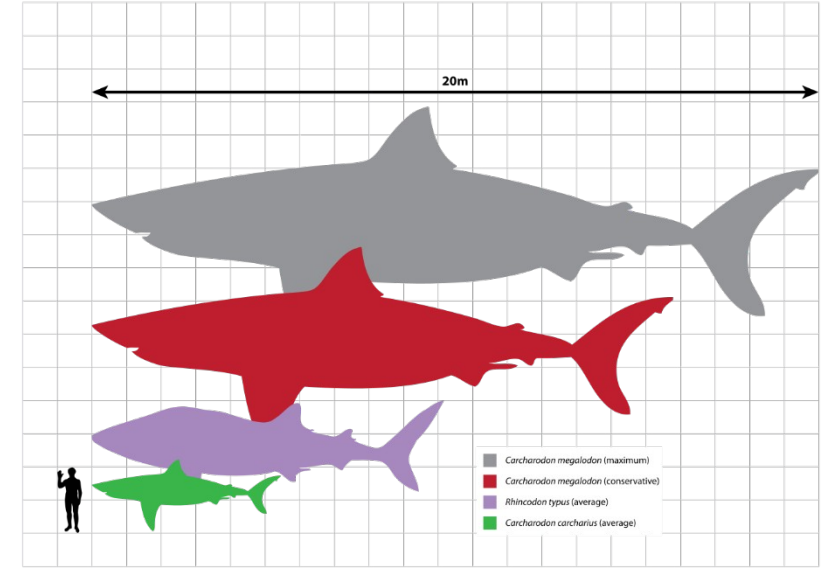
List



- An ordered sequence of elements, usually all of the same type
- Initialized like `[3, 1, 4, 1, 5, 9]`
- Empty would be `[]`
- Single element would be `[4]`
- Type of brackets matter – must be SQUARE
- Very important in data science!

List basics

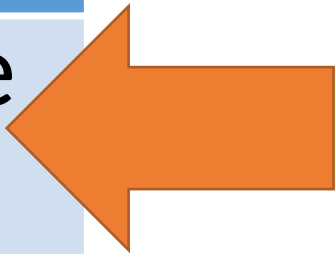
- Getting the length
 - `len(myList)`
 - Note indices start at 0, so the last index is `len(myList)-1`
- Getting the *i*th element
 - `myList[i]` $\leftarrow$ *i* is called an **index** (important term)
 - First element
 - `myList[0]`
 - Second element
 - `myList[1]`



Example

- `animals = ["Parrot", "Cow", "Horse"]`

<code>animals[0]</code>	<code>animals[1]</code>	<code>animals[2]</code>
Parrot	Cow	Horse



`animals[len(animals)-1]`

`len(animals)` is equal to 3

Looping through a list

Iterating through a list refers to walking through it element by element ← **Important term!**

Way 1:

- for item in myList:
 - # item is myList[0], then myList[1], then myList[2]...
- **Use this one if you don't need the index!**

Way 2:

- for i in range(len(myList)):
 - item = myList[i]
- **Use this one if you need the index!**



Adding to a list

- `myList = []`
 - `myList.append(3)`
 - `myList.append(4)`
 - Results in `[3,4]`
-
- Useful when building up a list from scratch
 - Often done in a loop (demo)



Checking if a list contains an item

- One way
 - Do a for loop, check every item with ==
 - Kind of annoying given how common this is
 - Python offers a way to make this nicer
- In keyword again!
- If 'a' in ['a', 'b', 'c']:
 - #This will happen
- If 'd' in ['a', 'b', 'c']:
 - #This will not happen
- If 'd' in ['a', 'bd', 'c']:
 - #This will also not happen
- For loop not usually good style for this





Strings are kind of like lists!

- Can index them:
 - `s = "hello"`
 - `print(s[0])` #prints h
- Can take their length
 - `s = "hello"`
 - `print(len(s))` #prints 5
 - *Useful for checking for invalid lines (small or 0 length after strip())*
- Can check if something is in them:
 - If "o" in "hello":
 - #this will happen
 - If "x" in "hello":
 - #this will not
- Cannot append, though

Splitting strings



- Strings by default are a list of characters
- Sometimes, that isn't what I'm looking for
 - Example: "james,fred,george"
- `myString = "a,b,c"`
- `myList = myString.split(",")`
 - `myList` is now `["a","b","c"]`
- Tip: `Split()` without anything in the parens will split on whitespace

Style tip with lists

- Often good to end variable name with an “s” (i.e. pluralized)
- `grades = [76, 83, 92, 86]`
- `for grade in grades:`
 - `print (“You got:”, grade)`

Lab out

- Practice with some of what we talked about today
- Reading data into a list, checking for errored values
- Next time: More with lists (slicing, etc.)